

THE FIELDWORK DECISION

The next useful check.

A citizen reports a stream concern. A coordinator has two hours. Rill helps decide what to check next and keeps the follow-up visible.

120 minutes available

The signature interaction: a reviewable fieldwork plan with reasons for each choice and explicit deferrals.

Primary track: Data-to-Insight

For river groups, citizen-science coordinators and research programmes.

Try the hosted app: rill-streams.web.app

A visible path after Submit

Rill joins observation, verification, human review and a dated recheck in one workspace.

foam		All sites	Priority first
All observations 8	Needs review 8	Reviewed 0	In action 0
OBSERVATION / LOCATION		VERIFICATION PRIORITY	STATUS
Surface foam Willow footbridge		High priority	Needs review
Surface foam Willow footbridge		High priority	Needs review
Surface foam Willow footbridge		High priority	Needs review
No visible concern · Wildlife seen Upstream gardens		Routine	Needs review
4 observations		Priority is a suggestion for follow-up, not a water safety score.	

Actual product screen. Synthetic demonstration observations.

01 A useful observation

Record place, time, visible signs and uncertainty. Preserve the source and notes.

02 A responsible decision

Inspect evidence separately from priority. A coordinator reviews and assigns the work.

03 A checked outcome

Record the task result, schedule a recheck and keep the conclusion accountable.

Evidence before confidence

Concern, evidence strength and work priority answer different questions. Rill keeps the distinctions visible.

A transparent two-hour plan

Versioned deterministic rules choose eligible checks within available minutes and expose the reasons. Restricted sites or serious signs need expert handling. Priority describes work to verify, never contamination probability. Time allowances are estimates. The core needs no trained model or paid AI key.

One Health with defined limits

Habitat change, animal contact and human use provide context. Appearance cannot establish pathogens or toxins. Citizen tasks stay on accessible public banks. Local advice and qualified judgment determine appropriate follow-up. [1]

A usable, inspectable system

Firebase Hosting serves the interface; Cloud Run uses PostgreSQL on Cloud SQL. Authenticated workspaces separate coordinator and volunteer roles. Decisions retain the factors and rule version used at review. Resolution requires a completed recheck after action.

Environmental data that can leave the app

The live OneAquaHealth directory provides names and coordinates only. CSV/JSON imports require explicit site matching. Export JSON, CSV, GeoJSON or experimental FHIR R4 with environmental locations and observations. No patient records or profile-certification claim. [2]

[1] CDC: How to Recognize a Harmful Algal Bloom

[2] HL7: FHIR R4 Observation. OneAquaHealth maintains an evolving draft IG.

Implementation and reproducible setup: github.com/shi1720/OneAquaHealth

A pilot that can prove value

The first buyer hypothesis is a coordinator running an existing river or catchment monitoring programme. Volunteers contribute free.

€149 proposed per programme / month

Managed coordination, hosting and support. Pricing has not been tested with customers.

The value and cost hypotheses

Buyer value: four hours saved at an assumed €40 per hour would free €160 of monthly capacity.
Delivery budget: €15 infrastructure + €30 support (45 minutes at €40/hour) + €5 operations = €50 per month. At €149 revenue, €99 remains before development, sales, taxes and fixed overhead. These are assumptions, not measured savings or margins.

Eight weeks, alongside an existing programme

Agree local protocols and reviewer responsibilities. Measure coordinator time, assignment delay, documented rechecks and safe coverage against the existing process. Redesign if the extra tool creates more work than it removes.

Current evidence: a hosted application with a tested workflow and synthetic demonstration observations. No customer, pilot agreement, measured ecological benefit or field-validated prediction is claimed.

Category context: Cartographer sells environmental monitoring subscriptions.

Complementary ecosystem: OneAquaHealth project solutions.